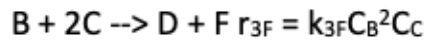
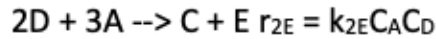
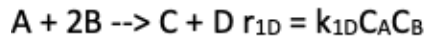


E09 Multiple Reactions in an Isothermal PFR

The following reactions are carried out in the gas phase in an isothermal PFR:



Where $k_{1D} = 0.25 \text{ L/mol} \cdot \text{min}$, $k_{2E} = 0.1 \text{ L/mol} \cdot \text{min}$, $k_{3F} = 5 \text{ L}^2/\text{mol}^2 \cdot \text{min}$, $C_{T0} = 0.4 \text{ mol/L}$, $V = 500 \text{ L}$, $y_{A0} = y_{B0}$, $u_0 = 100 \text{ L/min}$. Plot the species concentrations as functions of PFR volume.

to get $C_j(V)$, we need to solve the mole balances

$$\frac{dF_j}{dV} = r_{j,\text{net}}$$

to get $r_{j,\text{net}}$, we use the stoichiometry and rate matrices:

$$r = \begin{bmatrix} k_{1D} C_A C_B \\ k_{2E} C_A C_D \\ k_{3F} C_B^2 C_C \end{bmatrix}$$

reactions are already written s.t.

stoichiometric coefficients on D (rxn 1),

E (rxn 2), & F (rxn 3) are 1.

So, no need to re-write.

$$V = \begin{bmatrix} -1 & -2 & 1 & 1 & 0 & 0 \\ -3 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & -2 & 1 & 0 & 1 \end{bmatrix}$$